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A critical review of heart sound signal segmentation algorithms

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ABSTRACT

This paper discusses four heart sound segmentation (HSS) methods: Wavelet transform, Fractal decomposition, Hilbert Transform, and Shannon Energy Envelogram, in order to identify the different cardiac sounds. Many research studies related to heart signal analysis, have adopted these methods to give high heart sound segmentation results, especially, for the identification of first and second heart sounds and murmurs. Performance of the heart sound segmentation results have also been compared with one other to identify the most efficient method(s), and it has been found that Shannon energy envelogram provides the best accuracies among the segmentation methods. Understandings of the segmentation methods for heart sound may pave the way for more advanced studies in other heart-related researches, including heart sound classifications.

1. Introduction

A beating heart produces short but audible heart sounds and results in blood flow, for the transportation of oxygen as well as other important nutrients throughout the body. They are distinctively different from heart murmurs, which result from the turbulence of blood flow. Diastole and systole phases of the cardiac cycle of a human heart produce four (4) differentiable heart sounds, which are normally referred to as the S1, S2, S3, and S4 heart sounds. These sounds are produced due to the opening and closing of the heart valves. The S1 and S2 heart sounds are the two main audible heart sounds of each cardiac cycle, occurring at relatively higher frequencies and amplitude as compared to the S3 and S4 heart sounds.

Fig. 1 shows the structure of the human heart. The first heart sound (S1) occurs when the mitral and tricuspid valves are closed. Higher pressure on the left side of the heart results in the closure of the mitral valve making a louder sound than the sound of the closure of the tricuspid valve, and as such, the closure of the mitral valve acts as the main sound component of the first heart sound. The second heart sound (S2) can be heard due to the closure of aortic and pulmonic valves; occurring at the end of the ventricular systole and signifying the start of the diastole. Again, due to the higher pressure on the left side of the heart, the closure of the left aortic valve dominates the S2 heart sound. The third heart sound (S3) occurs when the mitral valve opens, causing the left ventricle of the heart to be filled with a large amount of blood. Generally, the pitch of the S2 heart sound is higher than the pitch of the S3 heart sound. The fourth heart sound (S4) occurs before the S1 heart sound in the late diastole when the atria contracts to force blood into the left ventricle, and it can be a significant sign of diastolic heart failure (Healio, 2017). The S1 and S2 sounds are considered as normal or fundamental heart sounds, while S3 and S4 sounds are two additional gallop sounds that may not be heard in a healthy adult heart. Hence, the S3 sound may not be a pathological heart sound in origin; thus it can be normal to hear from a child's heart (Vanderpool, 2009).

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Another heart sound that may be heard is a murmur, which produces blowing, whooshing, rasping, or swishing sounds from or near the heart. Heart murmurs occur when the heart valves are not properly closed or when the blood flows over a narrowed heart valve. These murmurs may not signify any symptoms of heart disease, however, loud murmurs or irregular sound variations may indicate an abnormal heart (Joshi, Tomar, & Tomar, 2014).

Heart sounds may be described by their intensity, pitch, location, quality, and timing during the cardiac cycle. Intensity may vary according to the amplitude of the heart sound signals. The S1 and S2 heart sounds have a comparatively higher pitch to murmurs. Depending on the heart sounds, murmurs may occur at the early, mid, or late systole or diastole (Healio, 2017). Generally, pitch, intensity, and cardiac sound time duration may be used to determine the location of the specific heart sounds. Fig. 2 illustrates a waveform representation of S1, S2, S3, and S4 sounds.

Heart sounds may be used for the diagnosis of some heart diseases, especially those that are related to blood circulation, and blood vessels, and may allow sufferers respite from heart attacks. Segmentation of the heart sounds and murmur is important to enable the automation of a heart sound classification system for accurate heart conditions prediction. This may assist physicians in determining the patient's heart condition as well as to prescribe appropriate early treatments to the patients.

A phonocardiogram (PCG) plots recordings of a heart sound and murmur during the cardiac cycle, and analysis of a PCG signal is useful in detecting abnormalities of a heart by using time-frequency analysis and classifications of the heart signals (Milani, Emeroylariffion, Silva, & Nanayakkara, 2021). PCG signal may be segmented according to the different components of a heart sound and murmurs, to enable analysis, with the energy of the PCG signal used to validate the cardiac cycle.

On the other hand, an electrocardiogram (ECG) records the electrical impulse of the heart muscles, and it may also be used to observe the conditions of a heart. Changes in the electrical charge of the heart impulses may be detected by using ECG sensors attached to the skin, with a graphical representation of heart impulses provided by showing the shape of its basic waves, namely its P, QRS, and T waves. These waves convey information on the electrical functionality of the heart. Amplitude and intervals of the P, QRS, and T waves of the ECG signal may then be used to detect abnormalities of the heart, through analysis of the electrical activities (Joshi et al., 2014).

Heart sound analysis system is composed of signal pre-processing, sound segmentation, feature extraction, and classification. Segmentation plays a major role in identifying normal and pathological events of the heart and is commonly based on segmentation and identification of the boundaries of the main cardiac sounds (S1, S2, S3, and S4) and murmurs, as well as separating heart sounds from noisy signals. Some researchers (Dissanayake, Denman, Sridharan, & Ghaemmaghami, 2020; Liu, Wu, & Ma, 2018) have specified that heart sound segmentation is a crucial step for feature extraction and classification, and to allow diagnosis of a patient's heart conditions. There are many computational heart sounds and murmur segmentation algorithms that have been developed, including Shannon-entropy, wavelet Transform, S-transform, Hilbert transform, logistic regression-hidden semi-Markov model, frequency-energy method, time-frequency analysis, dynamic clustering method, Shannon Energy Envelogram, fractal decomposition, Ensemble Empirical Mode Decomposition (EEMD), Matching pursuit algorithm, and deep learning-based algorithms. These methods mainly identify the locations and boundaries of each heart sound component and pathological murmurs, with promising results for both normal and abnormal PCG and ECG signals.

2. Segmentation methods

Many computer-aided heart sound segmentation methods have been proposed in the literature, including Hidden Markov Models (HMM), Hidden semi-Markov model (HSMM), Visual Molecular Dynamics (VMD), correlation coefficient matrices, and peak intervals pattern (Li et al., 2020). Other heart sound segmentation methods that use a combination of methods include Convolutional Neural Network (CNN) method based on HSMM (HSMM-CNN), and Gaussian Mixture Model (GMM) based HSMM (HSMM-GMM) (Li et al., 2020). Commonly utilised heart sound segmentation methods, such as Wavelet transform, Fractal decomposition, Hilbert envelope, and Shannon Energy Envelogram, operate in either the time-domain, frequency-domain, or combined time-frequency-domain, and

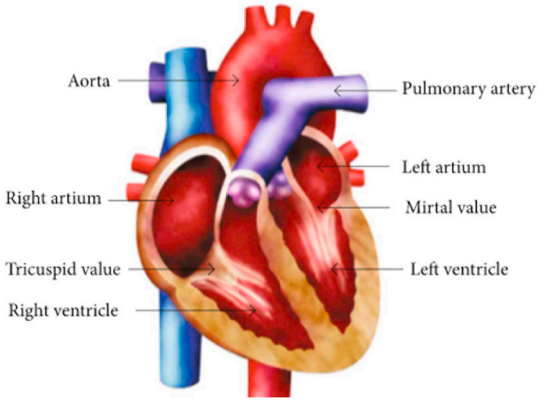


Fig. 1. Structure of the human heart, as of (Li, Li, Tang, & Xiong, 2020).

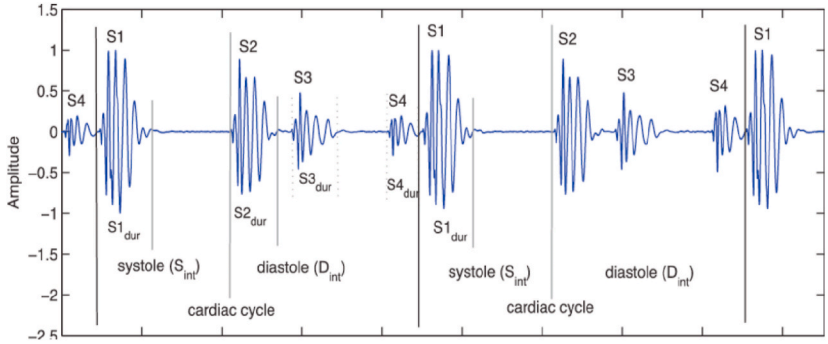


Fig. 2. Waveform representation of S1, S2, S3, and S3 sounds in systole and diastole intervals, as of (Varghees & Ramachandran, 2014).

they are commonly used as a basis for the development of more advanced segmentation methods. Some of these methods are developed by combining each other, such as combining peak intervals pattern with Hilbert envelope (Milani, Abas, & De Silva, 2019), and combination of the Wavelet transform and Teager Energy Operator (TEO) (Buza & Dizdar, 2017). This paper provides a thorough literature review on four (4) popular heart sounds and murmurs segmentation methods: (i) Wavelet transform, (ii) Fractal decomposition, (iii) Hilbert envelope and (iv) Shannon Energy Envelogram, and some other developed methods that have adapted these four (4) methods in heart sound segmentation process. Signal pre-processing, heart sound, and murmurs identification and classification methods that have been used by some early research studies, have also been briefly described.

2.1. Wavelet transform

Wavelet transformation is an efficient method for evaluating small waves. It may be applied in different applications, including data compression, noise removal, pattern recognition, and fast computation (Pathak, 2009); with the method used to change the time extension by shifting and scaling the signals. Wavelet is used to evaluate a given signal from a mother wavelet (Ψ) and a linear mixture of functions of expanded and shifted versions. This includes two transformation techniques: Continues Wavelet Transform (CWT) and Discrete Wavelet Transform (DWT). The CWT is commonly used to filter a large range of signal samples to obtain different scales and can be computed by:

$$(b,a) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} \Psi' \left(\frac{t-b}{a} \right) s(t) dt \tag{1}$$

where Ψ' represents the complex conjugate of Ψ, which is defined in the open (b,a) half-plane $(b \varepsilon R, a > 0)$. The parameters b and a represent the time shift and scale factor of the analysed wavelet, respectively (Yamamoto, 1994). On the other hand, DWT is primarily used to de-noise and compress noisy signals, with DWT computed by:

$$\Psi_{m,n}(t) = a_0^{-\frac{m}{2}} \Psi \left(\frac{t - nb_0}{a_0^m} \right) \tag{2}$$

where m and n are integer values, related to the scale and shift parameters a and b respectively, and discretised as $a = a_0^m$ and $b = nb_0$ (Yamamoto, 1994).

Wavelet Transformation can be applied to a window of cardiac cycles as input signal patterns to detect peaks of the signals. This segments specific cardiac sounds and murmurs by determining features of the heart signals. Wavelet theory provides an integrated outline and clear approach for many different ideas and methodologies that can be developed independently in numerous fields of study. Researchers (Huiying, Sakari, & Iiro, 1997, and Say, Dokur, & Olmez, 2002) have used wavelet transformation to segment heart sounds from noisy phonocardiogram (PCG) and Electrocardiography (ECG) signals, to determine different components of a heart sound and murmurs.

Wavelet transform is very popular and has been adopted to segment heart sound and murmurs from PCG signals, especially in segmenting the S1 and S2 sounds (Barschdorff, Femmer, & Trowitzsch, 1995). Thus, (Kumar, Carvalho, Antunes, Henriques, et al., 2006) have proposed to use wavelet decomposition with an adaptive threshold to differentiate between S1 and S2 heart sounds, and murmurs, whilst Castro et al. (2013) have used the wavelet analysis method for a similar purpose with 90.9% and 93.3% success rates in segmenting S1 and S2 sounds, respectively. Segmentation of S1 and S2 sounds of PCG signals has been further elaborated by Dokur et al. (2008), whereby a novel segmentation method based on the 3rd, 4th, and 5th decomposition-level detail coefficients of wavelet transformation has been proposed. Also, Dokur et al. (2008) have analysed the powers of detail coefficients of five sub-bands of the wavelet transform to enable the extraction of features from the segmented S1 and S2 sounds. Likewise (Say et al., 2002), have used wavelet detail and approximation coefficients of the 2nd and 6th decomposition levels to segment S1 and S2 heart sounds, whilst Liang, Lukkarinen, and Hartimo (1997) have proposed using wavelet decomposition and reconstruction method to segment S1 and S2 heart sounds, as well as the systole and diastole periods of both normal and abnormal PCG signals. Consequently, they have achieved

93% accuracy.

Baddour, Courtemanche, and Millette (2008) have combined both Mel-Frequency Cepstral Coefficient (MFCC) and Discrete Wavelet Transform (DWT) and introduced Mel-Scaled Wavelet Transform (MSWT) algorithm to segment S1 and S2 sounds from PCG signals. Similarly, Kouras, Boutana, and Benidir (2012) have used Shannon entropy with DWT to detect S1, S2, and S3 sounds, as well as some pathological murmurs of a PCG signal, by defining threshold values based on the entropy of detailed coefficients at each level of the DWT. Then Kumar, Carvalho, Antunes, Paiva, and Henriques (2011) have segmented S1 and S2 sounds in two (2) phases. In phase 1, S1, and S2 time limits in the wavelet domain are identified, whilst in phase 2, S1 and S2 sounds are differentiated from the high-frequency components of a PCG signal by adapting a threshold.

Wavelet transform has also been used as a filter, to refine heart sounds from noisy environments. Thus, (Wang, Kim, Ling, & Soh, 2005) have proposed to segment S1 heart sound from noisy environments using wavelet transformation. Similarly, (Kumar et al., 2007) have introduced an automatic method to discriminate the S1, S2, and S3 sound from murmurs and noise by using wavelet transform-simplicity filters on PCG signals of children and elderly heart patients. Morlet wavelet correlating filter has been used in (Rajan, Doraiswami, Stevenson, & Watrous, 1998), by utilising wavelet-based bank correlators to detect sound components of PCG signals and discriminate main cardiac sounds from non-heart sound components. Similarly, Tovar-Corona et al. (1998) have used Continuous Wavelet Transform (CWT) to segment systolic murmurs by analysing the rising and falling characteristics of murmurs, as well as their intensity variations, while (Santos's', 2001) has proposed a wavelet-based time-frequency analysis technique for the detection of S1 and S2 heart sounds in PCG signals.

In the past five years (2015–2020), some research studies (Jain & Tiwari, 2018; Patidar, Pachori, & Garg, 2015; Nivitha Varghees & Ramachandran, 2017; Meintjes, Lowe, & Legget, 2018; Golpaygani, Abolpour, Hassani, Bajelani, & Doyle, 2015; Nivitha Varghees & Ramachandran, 2017; and Buza & Dizdar, 2017) have used signal pre-processing and wavelet transformation based methods to segment heart sounds and murmurs from PCG signals that have been recorded in noisy environments. Hence, Jain et al. (2018) have used Tunable Quality Wavelet Transform (TQWT) method to segment normal and pathological heart sound signals from noisy PCG signals. Discrete Wavelet Transform (DWT), which includes Passband (P), transition band (T), and stopband (S), has been used to pre-process the input PCG signal. The authors then used the Fano factor to measure the distribution of the Fano noise to produce a signal with important fundamental heart sounds. The Fano factor was also used to select the desired signal with a low-level noise and adaptive threshold of a certain level to detect the peaks of the PCG signal. Envelop extraction method was then used to obtain the signal by using the normalized average Shannon energy (NASE). Similarly, the TQWT method has also been used in (Patidar et al., 2015) to reconstruct murmur-free heart sound signals, and to identify heartbeat cycles.

The study (Nivitha Varghees & Ramachandran, 2017) has proposed a combined method for PCG signal delineation and murmur classification without the use of reference signals to automatically detect and classify heart sounds and murmurs. The authors have used Empirical Wavelet Transform (EWT) method to investigate frequency components of heart sound signals, which have been pre-processed by computing their spectral boundaries. The Shannon entropy envelope extraction method was then used to reduce the effect of residual noise in an envelope to obtain S1 and S2 sounds according to the timings of the systole and diastole. The overall S1 and S2 segmentation accuracy of this method have been reported to be 91.92%. Then Meintjes et al. (2018) have used a low-pass Butterworth filter with 25 Hz cut-off frequency to filter the higher frequency noises of ECG signals, before passing it through a Continuous Wavelet Transform (CWT) to segment S1 and S2 heart sounds from the signal. The authors have also created scalograms from the segmented S1 and S2 sounds to analyse the time-frequency resolution of each segmented S1 and S2 sound. Likewise, Golpaygani et al. (2015) have used the 9th level discrete wavelet decomposition of the original heart sound signals to identify the coefficients of all sound components and reconstruct the signal with only the main cardiac sound frequencies by ignoring frequencies in the range of 10–400 Hz. The frequency spectrum of the reconstructed signal was then used to identify S1 and S2 heart sounds. Consequently, 93% S1 and S2 segmentation accuracy have been reported using the proposed WT method.

The WT is also commonly used as a filter to remove noises from PCG signals (Kumar et al., 2011). Thus, (Nivitha Varghees & Ramachandran, 2017) have used Stationary Wavelet Transform (SWT) method to remove low-frequency murmurs and high-frequency noises from input PCG signals. The filtered PCG signal is then enveloped using the Shannon entropy method. A defined amplitude threshold has been adopted on the enveloped signal to determine endpoints of the S1, S2, S3, and S4 heart sounds. The proposed segmentation method has reported 93.53% overall accuracy in distinguishing between S1 and S2 sounds in noisy signal conditions. Also, Buza et al. (2017) have proposed a combined Wavelet Transform and Teager Energy Operator (TEO) method to segment S1, S2, S3, and S4 heart sounds from PCG signals. Specifically, the TEO method with WT has been used to define a non-linear energy tracking operator to estimate the amplitude and instantaneous frequency of a PCG signal. Overall segmentation accuracy of the proposed method has been reported as 88%.

2.2. *Fractal decomposition*

Fractals are self-similar patterns, which can be considered as a mathematical set. Fractal dimension can be used to describe and analyse the geometry of linear figures. It is sensitive to the information contained in the sequence of a PCG signal, and aids in improving signal classifications. Also, the method works well in characterising and analysing heart signals (Verma, Kumar Biswas, Potdar, & Associate, 2013). Heart sounds are fractal but they are not self-crossed, and thus, a scaling factor is required to scale each axis of a heart sound signal. Gnitecki et al. (2003) have suggested Variance Fractal Dimension (VFD) identify the locations of the fundamental heart sounds using an adaptive threshold. The VFD may provide better performance in segmenting heart sounds in the presence of respiratory noises (Gavrovska, Zajić, Bogdanović, Reljin, & Reljin, 2017). The VFD method can be used to segment S1 and S2 sounds, as demonstrated in Ishida et al. (2007) for the segmentation of ECG signals. Then Verma et al. (2013) have also used VFD to identify the

boundaries of S1 and S2 sounds from the QRS waves of ECG signals. Similarly, Gavrovska et al. (2017) have proposed a method of segmenting S1 and S2 heart sounds to allow an analysis of their structure and patterns using fractal dimensions. Higuchi's algorithm is also a popular method that can be used to calculate fractal dimensions of PCG signals. The algorithm has been proposed in (Juniati, Khotimah, Wardani, & Budayasa, 2018) to identify S1 and S2 sounds, and classify normal and abnormal PCG signals using k-nearest neighbours (KNN) and fuzzy c-mean clustering methods. Similarly, Mukherjee, Pathak, and Roy (2014) have proposed a method using Higuchi's fractal dimension algorithm to differentiate between normal S1 and S2 sounds from heart sounds that have been recorded from valvular dysfunction hearts.

During the past five years, some studies, including. (Azmy & Mohamady, 2017; Koutsiana, Hadjileontiadis, Chouvarda, & Khandoker, 2017; Koutsiana et al., 2017; and Thomas, Hsi, Boon, & Gunawan, 2016) have used fractal dimension technique along with signal pre-processing algorithms to develop a better heart sound and murmur segmentation algorithm. Azmy et al. (2017) have proposed a low-pass Butterworth filter to remove noises from PCG signals. Multi-Fractal De-trended Fluctuation Analysis (MFDFA) and VFD methods are used for the heart sound segmentation and feature extraction process. The VFD is calculated using:

$$var(s(t_1) - s(t_2)) = \left|t_2 - t_1\right|^{2H} \tag{3}$$

where s is the input PCG signal, t_1 is the first measured time of the signal, t_2 is the delayed time after t_1, and H is the Hurst exponent that is used to smoothen the signal. Furthermore, Azmy et al. (2017) have classified the extracted heart sound features into normal and abnormal classes using the Support Vector Machine (SVM) technique and achieved 96.875% overall classification accuracy.

Koutsiana et al. (2017) have examined the transferability of Wavelet Transform (WT) and Fractal Dimension (FD) methods to analyse the locations of the Fundamental Heart Sounds (FHS). The time intervals between the sound peaks are calculated by:

$$S(2i+1) - S(2i) < S(2i+3) - S(2i+2) \tag{4}$$

where $S(1)$ is a vector that has been created from a user-defined binary threshold, and the locations of start and endpoints of every heartbeat, $i = 1, \ldots, \left(\frac{N_1}{2}\right) - 2$, with N_1 representing the length of $S(1)$. The authors have assumed the longer time interval as a diastole interval, whilst the shorter time interval has been taken as the systole interval. Consequently, the S1 and S2 sounds can be identified by considering the calculated systole and diastole intervals; with the corresponding sound peak within the detected diastole and systole intervals defined as S2 and S1 sounds, respectively. Using the proposed WT-FD method, 89% segmentation accuracy is achievable. Then Koutsiana et al. (2017) have also proposed a WT-FD method to segment fatal heart sounds from superimposed noise with amplitudes of the PCG signals that contain stationary and non-stationary background noises used as a basis for the proposed WT-FD segmentation method. However, Thomas et al. (2016) have used PCG signals from two different heart sound databases and filtered them using a Butterworth filter with a 10,800Hz cut-off frequency. Fractal decomposition is then applied to the filtered signal to identify the start and endpoints of the S1 and S2 sounds. The overall segmentation accuracy on heart sounds from the first database is 94.22%, whilst the accuracy is 96.58% from the second database.

2.3. Hilbert envelope algorithm

Hilbert transform is one of the main linear operators in signal processing. It is bounded on $L^p(R)$ for $1 < p < \infty$ to take functions from a real variable, and produces an analytic function (Hamidi, Ghassemian, & Imani, 2018). The results from the Hilbert transform can be generalized for a singular integral operator in several dimensions. This method is used to decompose non-stationary and non-linear signals, and extract instantaneous frequencies from the signal (Klingspor, 2015). Hilbert transform is also used to create a time function from another time function that contains the same characteristics, but with different signal shapes. Also, this method can provide an energy density spectrum, i.e. energy distribution of a signal in the frequency domain. Hilbert transform acts as a signal transformation operator that has induced a vast number of applications, including; sampling narrowband signals, medical imaging, and system response analysis, where signal theory has been applied. Hilbert transform can be calculated as:

$$H(u(t)) = \lim_{\in \to 0} \frac{1}{\pi} \int_{|s-t|>\in} \frac{u(s)}{t-s} ds \tag{5}$$

where $u(t)$ is the input function, and the Hilbert transform of $u(t)$ is obtained by providing $(-\pi/2)$ phase shift to every frequency component present in $u(t)$ (Thomas et al., 2016).

There are many heart sound segmentation applications which have been based on the Hilbert envelope algorithm. Among them, Xu et al. (2010) have proposed the Wavelet-Hilbert-Huang transform method to envelop abnormal heart sound signals to identify different heat diseases, with the enveloped signal used to separate S1 and S2 sounds from a noisy signal. The authors have achieved 97% accuracy using the proposed method. Also, Chen et al. (2013) have used Hilbert-Huang transform to extract the envelope of a PCG signal, to allow segmentation of S1 and S2 sounds, whilst (Tseng, Ko, & Jaw, 2012) have used the same enveloping method to segment S3 and S4 sounds from ECG signals. On the other hand (Mondal, Bhattacharya, & Saha, 2013), have proposed an algorithm to segment S1, S2, S3, and S4 sounds using Hilbert transform and Heron's formula. An automatic heart sound segmentation and peak detection method have been proposed in (Sun, Jiang, Wang, & Fang, 2014), whereby Hilbert transform has been used to envelop the input signal to give average accuracies of 100% and 96.69% in S1 and S2 sound segmentation, respectively. Then Zhong, Guo, Ji, and Ding (2011) have also used Hilbert transform to envelope PCG signals and to differentiate S1 and S2 sounds from low and high noisy environments. They have demonstrated 94.3% sensitivity using the proposed method.

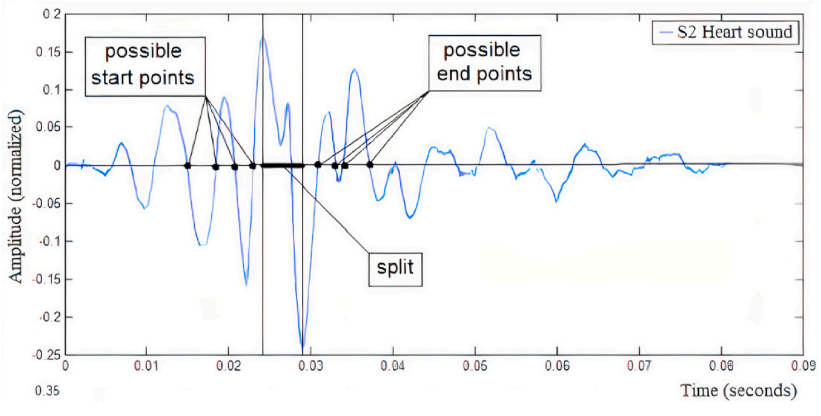
The signal enveloping process using the Hilbert transform is also known as the Hilbert enveloping algorithm. Therefore (Chourasia, Tiwari, & Gangopadhyay, 2014), and (Varghees & Ramachandran, 2014) have applied the algorithm to segment S1 and S2 sounds as well as systole and diastole intervals. Based on their segmentation results, the authors were able to identify the boundaries of fatal heart sounds. Likewise, Sun, Wang, Jiang, Fang, and Tao (2014) have segmented S1 and S2 sounds along with the systole and diastole intervals using a moving windowed Hilbert transform (MWHT) enveloping method. Similarly, Latif, Usman, Rana, and Qadir (2018) have used Hilbert envelop to extract features of PCG signals, to identify the states of S1 and S2 sounds. Along with main cardiac sounds, some murmurs of PCG signals have also been successfully identified using the Hilbert enveloping method. The method has also been used in (Pasteur & Koch, 1941) to segment S1 and S2 sounds in the presence of murmurs in the PCG signals to identify patients with lonescu-Shiley aortic bioprostheses.

The authors (Cherif & Debbal, 2016) have proposed to use a low-pass Finite Impulse Response (FIR) filter to filter the input PCG signals, with the filtered signal subsequently enveloped using Hilbert transformation to identify the time intervals between the S1 and S2 heart sounds. Fig. 3 shows the Hilbert envelope of an S2 heart sound.

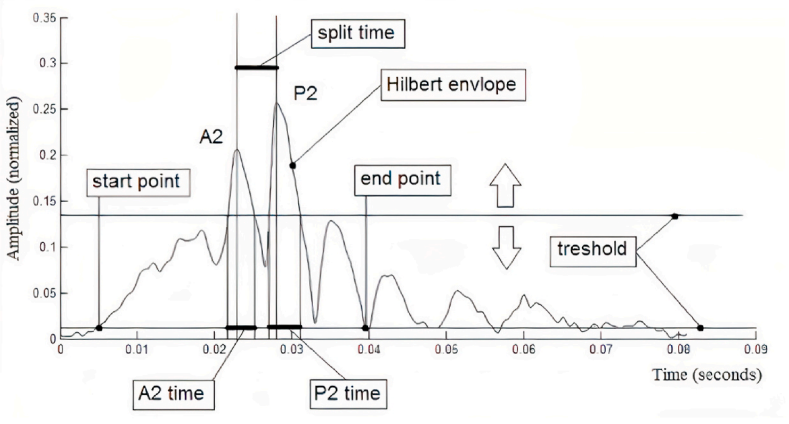
From Fig. 3, the split time between the aortic component (A2) and pulmonary component (P2) can be calculated. The same Hilbert envelope calculation is also performed to the S1 sound to identify the time interval between the mitral component (M1) and tricuspid component (T1).

The authors (Saini, Nath, Tripathi, Saini, & K.K, 2017) have proposed a method to analyse details of PCG signals that have been recorded from people before and after physical exercises. Hilbert-Huang Transform (HHT), which includes a two-steps process: Empirical Mode Decomposition (EMD) and Hilbert Spectral Analysis (HSA) is then used; with the EMD method that used to decompose the original PCG signal into Intrinsic Mode Function (IMF), and the HSA applied to the Hilbert transform on different intrinsic mode functions. Fig. 3 shows the result obtained in (Saini et al., 2017). Hilbert spectrum can also be used to find and analyse the frequencies of a signal. Fig. 4(a) depicts the Hilbert spectrum of the PCG signal that has been recorded before physical exercise, whilst Fig. 4(b) depicts the Hilbert spectrum of the PCG signal that has been recorded after physical exercise.

In Fig. 4, the colour of the frequency component describes the energy of the input signal. The frequency components with 'red'

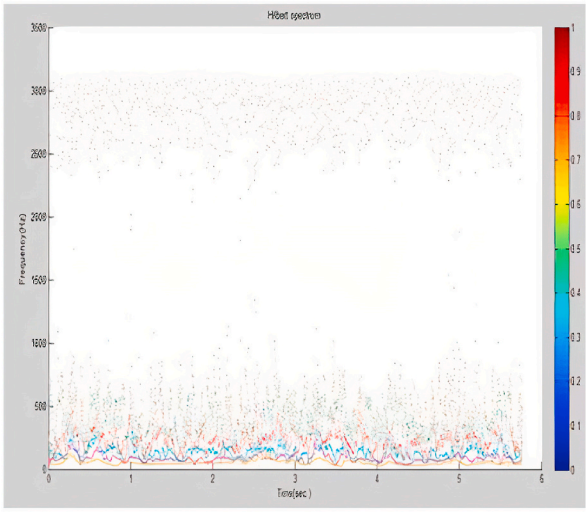


(a)

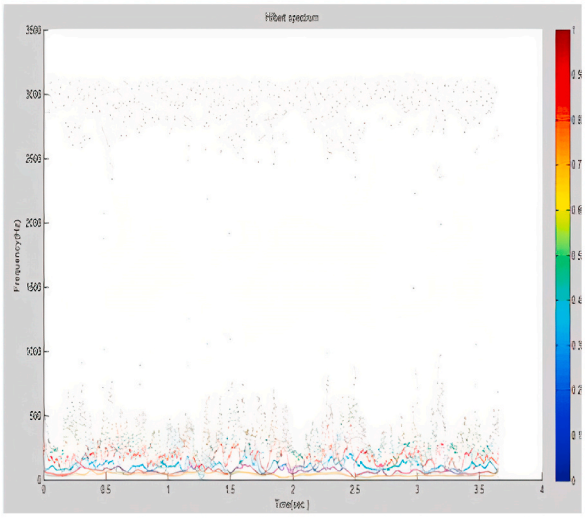


(b)

Fig. 3. (a) S2 heart sound of a normal PCG signal, (b) Hilbert envelope S2 heart sound of normal PCG signal of a) (A2 = Aortic component, P2 = Pulmonary component, as of (Cherif & Debbal, 2016).



(a)



(b)

Fig. 4. (a) Hilbert Spectrum of a PCG signal of person before exercises, (b) Hilbert Spectrum of a PCG signal of a person after exercises, as of (Saini et al., 2017).

colour give higher energies, whilst the 'blue' colour frequency components represent energies that provide less contribution to the PCG signal. The authors have concluded that the effect of frequency changes when comparing Fig. 4(a) and Fig. 4(b), maybe due to the respiration signal, and muscle noises that may have interfered with the heart sound signals.

The authors (Nivitha Varghees & Ramachandran, 2015) have proposed a method to discriminate heart sounds from murmurs. Two sub-bands: 1) low frequency (LF) to identify S1, S2, S3, and S4 sounds, and 2) high frequency (HF) to identify heart murmurs have been defined upon which Stationary Wavelet Transform (SWT) has been used to remove low-frequency artifacts and background noises as well as to decompose the PCG signal. The decomposed signal is then passed through a Hilbert envelope to determine real peaks and boundaries of the cardiac sounds and murmurs that are present in the LF and HF component signals. The proposed heart sounds and murmur differentiation method has been shown to give overall accuracy of 82.76%.

Moreover, Sharma (2015) has used Hilbert transformation to construct multiscale envelopes of PCG signals. The enveloped signal is then passed through a low-pass filter, which is then used to identify the boundaries of the S1 and S2 sounds. Power frequencies of both S1 and S2 sounds are then analysed to observe if they carry major energies. Likewise, Sharma (2016) has also used Hilbert transform to compute a smooth envelope of PCG signals to identify suitable boundaries of S1 and S2 sounds.

Consequently, Milani et al. (2019) have applied a low-pass FIR filter to filter out noises from PCG signals. Negative frequency components have been removed from the PCG signals, with the positive frequency components enveloped using an N-tap Hilbert filter.

Peaks of the enveloped signal have been detected via a defined threshold, which is based on the signal's amplitude. Prominent peaks of the Hilbert enveloped signal may then be segmented and labelled as S1 and S2 by calculating the distance between the identified prominent peaks. The shorter duration between peaks is considered systole, whilst the longer duration is as diastole. Subsequently, a prominent peak standing before systole is labelled as S1, whilst a prominent peak standing before diastole is labelled as S2.

Other than the use of Hilbert transform on PCG signals, some studies have also shown encouraging results by applying Hilbert transform on ECG signals. In their study, Quiceno-Manrique, Godino-Llorente, Blanco-Velasco, and Castellanos-Dominguez (2010) have applied Hilbert transform with an adaptive threshold to find the QRS complex and the location of R peaks of ECG signals. Similarly, Wood et al. (1994) have applied the Hilbert envelope to ECG signals and used it to segment the first heart sound.

Indeed, some researchers have successfully used Hilbert envelop algorithm to achieve higher heart sound segmentation accuracies (Jafari Tadi et al., 2016; Cherif & Debbal, 2016; Saini et al., 2017; Nivitha Varghees & Ramachandran, 2015; Sharma, 2015; Sharma, 2016; and Milani et al., 2019). Hence, Jafari Tadi et al. (2016) have proposed a real-time application to estimate beat-to-beat time intervals and heart rate, with a 3 dB bandpass 4th order Butterworth IIR filter with 1Hz and 45Hz cut-off frequencies to filter white noise from the input Seismocardiography (SCG) signal. The Hilbert adaptive beat identification technique (HABIT), which is based on the Hilbert transformation has been developed to identify the S1 and S2 sounds and calculate the required beat-to-beat intervals of the filtered SCG signal.

2.4. Shannon energy envelogram

Shannon energy (*SE*) highlights the medium-intensity signals whilst attenuating the effect of low-intensity signals (Sharma, 2015). It also can absorb both high-intensity and low-amplitude fluctuations in their magnitude (Kumar Sharma, Saha, & Kumari, 2014). This calculation method is commonly used to compute the envelopes of the signal by adaptive windows in an advanced way (Ali, Khan, Ahmad, & Khan, 2007b). Instantaneous Shannon energy can be calculated by:

$$SE\,(t) = -signal^2(t) \times \log\left(signal^2(t)\right) \tag{6}$$

Average Shannon energy within a specific time duration can then be calculated to evaluate envelope value (Meintjes et al., 2018), where N is the window size in a number of samples.

$$E_{avg} = -\frac{1}{N}\sum_{t=1}^{N} Shannon\ Energy\,(t) \tag{7}$$

The average Shannon energy can be standardized or normalized by:

$$E_{\eta} = \frac{E-\mu}{\sigma} \tag{8}$$

where E is the Shannon energy, μ is the average value of energy E, σ is the standard deviation of energy E of the signal, and E_n is the average Shannon energy standardized or normalized (Kumar Sharma et al., 2014).

The Shannon energy envelogram method has been used in many heart sound segmentation work. In his study (Nazeran, 2007), has implemented Shannon energy to differentiate S1 and S2 heart sounds from PCG signals. Also, Gill, Gavrieli, and Intrator (2005) have calculated Shannon energy of a homomorphic envelogram to identify S1 and S2 sounds from PCG signals, while Gamero et al. (2003) have enveloped both normal and abnormal PCG signals and subsequently, applied Shannon energy-based formulas to improve signal pre-processing as well as to reduce signal noises. Additionally, the boundaries of each S1 and S2 sound can also be detected from the extracted enveloped signal. Then (Liang et al., 1997) have achieved 93% segmentation accuracy by applying average Shannon energy in PCG signals to segment S1 and S2 sounds along with systole and diastole periods. Likewise, Ali et al. (2007) have proposed the S1 and S2 heart sounds segmentation technique by calculating the Shannon energy of PCG signals using adaptive windows and then extracting features using Shannon energy envelogram.

Shannon energy envelope has been used in heart sound peak identification (Jang Hyun Beak, 2009). An automatic heart sound peak detection algorithm using the Shannon energy envelope to detect S1 and S2 sounds has been proposed (Jang Hyun Beak, 2009). The proposed method has been shown to obtain 91.5% accuracy. Integration of the Shannon energy method with other segmentation methods has also been experimented in (Misal & Sinha 2012), whereby the authors have used Shannon energy envelope with Discrete Wavelet Transform (DWT) to segment S1 and S2 sounds from PCG signals.

The authors (Othman & Khaleel, 2017) have calculated the Shannon energy envelope of input PCG signals to identify the medium intensity of the signals and to remove the noise. A threshold is then set by observing the maximum energy value of the envelope to differentiate the S1 and S2 peaks and murmurs. Finally, the frequencies of segmented heart sounds and murmurs are used to classify the normal and abnormal PCG signals. Then Kang, Doroshow, McConnaughey, Khandoker, and Shekhar (2016) have proposed a 40–500Hz bandpass filter to eliminate low and high-frequency noises in PCG signals. The filtered signal is then enveloped using Shannon energy envelogram with 20ms sliding windows and 10ms overlap between successive windows. A threshold of 250ms; which

is based on a time interval between the identified sound lobes is set to isolate the potential candidates of S1 and S2 sounds. The time interval between each sound lobe is measured individually to capture the actual S1 and S2 sound lobes with a starting sound lobe of the longest time interval (diastole) defined as the S2 sound, and a starting sound lobe of the shortest time interval (systole) defined as the S1 sound. Additionally, the sensitivity and Positive Predictive Value (PPV) of the proposed heart sound segmentation algorithm has been calculated, and it can achieve >99.2% accuracy for both normal and abnormal PCG signals.

Moreover, Balili, Sobrepeña, and Naval (2016) have applied for the sixth order Daubechies wavelet filter to decompose and reconstruct original PCG signals. Shannon energy of the reconstructed signal is calculated by splitting the signal into 20ms frames with 10ms frame overlap, to calculate the average Shannon energy of each frame using:

$$E_s = -\frac{1}{N}\sum x^2_{norm}(i)\log x^2_{norm}(i) \tag{9}$$

Where x_{norm} is the normalized signal to the maximum value of the reconstructed signal, N is the number of samples in the current frame. Shannon energy per frame has also been computed:

$$E_s = \frac{E_s(f) - ME_s}{S(E_s)} \tag{10}$$

where ME_s is the mean of Shannon energy per frame $E_s(f)$, and SE_s is the standard deviation of E_s. The normalized Shannon energy of each frame is then used to identify the energies of the signal peaks, with the distance between peaks calculated to define a threshold to detect S1 and S2 sounds; which contain the highest energies among other heart sounds. Finally, the authors have compared the proposed segmentation method with previous research studies and showed that the segmentation error of their method is comparably lower than the previous studies.

The authors (Banerjee, Mishra, & Mukherjee, 2016) have used Empirical Mode Decomposition (EMD) based de-noising method to de-noise PCG signals. The de-noised signals are enveloped using Shannon energy to analyse intensities and the first derivative of the enveloped signal with local maxima and minima identified through the computation of intensities and first derivative to segment S1, S2, S3, and S4 cardiac sounds. The accuracy of the detected cardiac sound peaks and boundaries are evaluated by Heron's formula. The proposed heart sound segmentation method is robust and very accurate in detecting heart sound components in both normal and abnormal PCG signals.

Downsampling PCG signal into 2000Hz using Nyquist-Shannon's sampling theorem has been demonstrated in (Hamidah, Saputra, Mengko, Mengko, & Anggoro, 2017). The downsampled signal is passed through a low-pass filter with a 400Hz cut-off frequency to reduce the high-frequency noises. The Shannon energy envelope of the noise-free signal is then passed through a Finite Impulse Response (FIR) filter with a 25Hz cut-off frequency to smooth the enveloped signal. Both cardiac cycle time and cardiac number are then calculated by the autocorrelation of the enveloped signal. The S1, S2, S3, and S4 sound peaks of the enveloped signal are segmented according to the peak durations and the frequency with; S1: 100–160ms with 10–140Hz, S2: 80–140ms with 10–400Hz, S3: 40–80ms with 15–60Hz, S4: 30–60ms with 15–45Hz.

Banerjee et al. (2016) have proposed a median filter followed by a low-pass filter with 800Hz cut-off frequency to filter the high-frequency noises from input PCG signals. The filtered signals are passed through Variational Mode Decomposition (VMD) algorithm to decompose the signal into seven (7) modes with error variation of the signal analysed to identify its minimum error variation. The 3rd mode among all 7 modes has been observed to carry the minimum error. Shannon energy of the 3rd mode is then calculated to identify the S1 and S2 sound locations by determining threshold values to the signal. The segmentation accuracy of the proposed method has been shown to reach nearly 90% with less than 5ms error.

Shannon energy envelope is also popular in ECG signal processing. Then Kumar, Carvalho, Antunes, Henriques, et al. (2006) have used Fast Wavelet Transform (FWT) and Shannon energy to identify the sound lobes of the S1 and S2 heart sounds of ECG signals, and have achieved a 94.77% sensitivity rate and 96.16% specificity rate. Also, Choi et al. (2008) have generated Shannon energy envelope curves based on the timings of S1 and S2 sounds and examined the performance of the applied Shannon energy envelogram method in heart sound identification. On the other hand, (Saini, 2016) has used a three-degree normalized average Shannon energy calculation, as in equation (11) to segment:

$$E_s = -\frac{1}{N}\sum_{i=1}^{N} x^3(i) \times log x^3(i) \tag{11}$$

where $x(i)$ is the original signal, and E_s is normalized average Shannon energy. The calculated Shannon energy envelope with a defined soft threshold has been used to obtain the positions of S1 and S2 sounds by constructing a time gate between systole and diastole. The identified S1 and S2 locations are then linked with the number of R peaks in the ECG signal and have been shown to achieve a 95.6% success rate (Table 1).

Table 1
Summary of reviewed segmentation methods.

Segmentation method	Reference	High-frequency noise reduction	Remark
Wavelet transform	(Jain & Tiwari, 2018)	Discrete Wavelet Transform (DWT)	In this study, the authors have used heart sound series produced by the Texas Heart Institute at St. Luke's Episcopal Hospital, which contains both normal and pathological PCG signals. They have applied simulated noises, including white Gaussian noise, pink noise, and red noise, to test the performance of the proposed segmentation method. In their proposed method, they have applied a Tunable Quality Wavelet Transformation (TQWT) to decompose the signal up to twenty levels. The decomposed signal is passed through a threshold which is obtained according to the present noise level of the signal to identify the peaks of the fundamental heart sounds. The authors have compared their proposed segmentation method with a few past research studies which have used WT and constrained TQWT (CTQWT) methods, and have proven that their method can give higher accuracies in both noisy and non-noisy environments.
	Patidar et al. (2015)	TQWT parameters to remove murmurs from heart sound signals.	The authors have formed a global heart sound dataset, by combining five (5) different available heart sound datasets from Texas university, Michigan University, Loyola University, University of Washington, and the ALDMD clinical cardiology tools from Hennepin county medical center. They have considered four (4) classes, including normal, septal defect, valvular defect, and other defects. These classes contain a sample size of 17 in normal, 15 in septal defect, 101 in valvular defect, and 30 in other defects. The datasets contain many background noises and sampling frequencies that may be different in each chosen dataset. The signals are filtered via the TQWT method and then the filtered signal is enveloped using Cardiac Sound Characteristic Waveform (CSCW) analysis and timing information of the enveloped signal is used to identify S1 and S2 sound. After the segmentation process, the authors have classified the segmented signals with Least Squares Support Vector Machine (LS-SVM) based classifier with Radial Basis Function (RBF) kernel into the selected four (4) classes and achieved over 90% classification accuracy.
	(Nivitha Varghees & Ramachandran, 2017)	Empirical Wavelet Transform (EWT) for signal pre-processing	The PCG signals from eGeneralMedical, Michigan heart sound and murmur library, and bioscience heart sound databases, have been used; with the EWT method used to investigate the frequency components and spectral boundaries of the heart sound signals. The timings of systole and diastole are then calculated to identify S1 and S2 sounds. They have applied this method to test PCG signals of normal, split, click, murmurs of early-systolic, late-systolic, mid-systolic and pan- systolic, mitral and tricuspid stenosis, aortic and pulmonic regurgitation, fixed and wide S2 split, S1 split, mitral prolapse, ejection murmur and click, and different time-varying systolic and diastolic murmurs. The proposed method achieves over 90% accuracy in early-systolic and mid-diastolic segmentation, while mid-systolic, early-diastolic, late-diastolic, and click sounds segmentations give nearly 90% accuracy. On the other hand, pan-diastolic, and continuous segmentation give 100% segmentation accuracy.
	(Meintjes et al., 2018)	Low-pass Butterworth filter with 25Hz cutoff frequency	The authors have adapted 120 patients' heart sound recordings from the Physionet publicly available dataset and applied Continuous Wavelet Transform (CWT) to segment S1 and S2 sounds. Frequencies of the segmented S1 and S2 sounds are further analysed by extracting their scalogram. Scalogram features of the segmented S1 and S2 sounds have been used to train a CNN classifier; with another 15 heart sound signals used to test the trained classifier. The authors have reported 86% accuracy from the proposed CNN classifier, which is better as compared to other classifiers, including SVM and KNN.
	(Golpaygani et al., 2015)	N/A	The study has been conducted using heart sound data that have been collected from 100 cases at the Pediatric Clinic of Modares Hospital in Tehran, between 2010 and 2011. This dataset contains 8 normal heart sounds with an age range between 1–26, and the remaining 92 cases have a history of heart murmurs and other kinds of diseases. The 9th level discrete wavelet decomposition is used to identify the coefficients of all sound components and to reconstruct the signal. The S1 and S2 sounds have been categorised by

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Table 1 (*continued*)

Segmentation method	Reference	High-frequency noise reduction	Remark
			analysing the spectrum of the reconstructed signal and achieving 93% segmentation accuracy.
	(Nivitha Varghees & Ramachandran, 2017)	Stationary Wavelet Transform (SWT)	The authors have simultaneously recorded PCG signals of both normal and murmurs and used them to demonstrate the proposed two-channel heart sound segmentation framework. This framework has an amplitude-based threshold which is introduced to the filtered signal to segment S1, S2, S3, and S4 sounds, with 93.53% overall segmentation accuracy achieved.
	(Buza & Dizdar, 2017)	N/A	This experiment utilised 64 PCG signals, including 63 abnormal signals. Out of these 63 signals, 21 of them contain either S3 or S4 sounds or both of them. These PCG signals have been obtained from the open heart sounds database. A combination of the Wavelet transform and Teager Energy Operator (TEO) method has been implemented to segment S1, S2, S3, and S4 heart sounds of PCG signals. The overall segmentation accuracy from this method is 88%.
Fractal decomposition	(Azmy & Mohamady, 2017)	Low-pass Butterworth filter	The PhysioNet heart sounds dataset is considered in this study. The authors selected 200 heart sounds of both healthy individuals and individuals with a variety of heart diseases, including coronary artery disease, and heart valve disease. Multi Fractal De-Trended Fluctuation Analysis (MFDFA) and Variance Fractal Dimension (VFD) methods have been proposed to identify the different specifications of the heart sound signals. After applying the MFDFA method, the authors have conducted a classification process using the SVM method to classify normal and abnormal (heart sounds with various diseases) and achieved a 96.875% recognition rate.
	Koutsiana et al. (2017)	N/A	This study has used twelve fetal phonocardiograms (fPCG) signals to examine the S1 and S2 sounds of different heart rate conditions, including normal heart rate, tachycardia, bradycardia, and arrhythmia. The transferability of Wavelet Transform (WT) and Fractal Dimension (FD) methods have been used to segment S1 and S2 sounds and achieved 89% overall segmentation accuracy. Apart from this method, the authors have also assessed Teager Energy Operator (TEO) on another dataset, which contains different noises including environmental noise, maternal and fetal noise. The results from both methods have been compared with each other, with both methods giving comparable results.
	Koutsiana et al. (2017)	N/A	This study is an extension of (Elisavet Koutsiana et al., 2017). The authors have used a combined WT and FD method to emphasize heart sounds from superimposed noise and experimented with this method on 12 simulated fetal PCG (fPCG) signals, and the fPCG signals from the PhysioBank dataset, separately. The results have shown that the efficiency of locating the S1 sounds in the 12 simulated fPCGs, that contain the conditions of tachycardia, bradycardia, and arrhythmia, is slightly lower as compared to the detection of S2 sounds. The proposed segmentation method works well with both fPCG datasets, with the chosen PhysioBank dataset providing 81% mean accuracy in detecting the sound locations of S1 and S2 sounds.
	(Thomas et al., 2016)	Butterworth filter with a 10,800Hz cut-off frequency	Fractal decomposition is applied to the filtered signal to identify the start and endpoints of S1 and S2 sounds. The proposed method has been tested on 6 Mitral Regurgitation (MR) PCG signals that have been collected from the National University Hospital (NUH), Singapore, and 23 PCG signals from the University of Michigan Heart Sound and Murmur Library. The mean prediction power for the PCG signals collected from NUH is 94.22%, whilst the mean prediction power for the Michigan dataset is 96.58%.
Hilbert envelope	(Jafari Tadi et al., 2016)	3 dB bandpass 4th order Butterworth IIR filter with 1Hz and 45Hz cut-off frequency	The authors have used seismocardiogram (SCG) signals to observe the motion artifacts of heart sound signals. Before recording the signals, 30 participants were asked to be as motionless as possible, to get the best motion artifact signals from three different positions including supine, left side, and right side. Then, the Hilbert adaptive beat identification technique (HABIT), which is based on the Hilbert transformation, has been applied to the recorded signals to identify the S1 and S2 sounds. The authors have further developed the HABIT to classify the segmented signals into two categories, including motionless and motion artifacts, and achieved more than 97% average accuracy in detecting the motionless artifacts. The

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Table 1 (*continued*)

Segmentation method	Reference	High-frequency noise reduction	Remark
			average accuracy in classifying the motion artifacts on the supine left side, and right side positions is over 90% of the positive predictive value (PPV).
	(Cherif & Debbal, 2016)	Low-pass FIR filter	The PCG signals from 1 normal heart, 5 from mitral stenosis (MS), 4 from mitral regurgitation (MR), 1 from aortic regurgitation (AR), 4 from aortic stenosis (AS), and 2 from cardiac defects (ASD: Atrial Septal Defect), have been considered in this study. The PCG signals were recorded at the Boston Children's Hospital, Boston, USA. Since the recorded PCG signals contain noises, the authors have applied a low-pass FIR filter to filter the noise. The filtered signal is enveloped using Hilbert transformation to identify the time intervals between the S1 and S2 heart sounds. Also, the duration between the segmented sounds and the pulmonary artery systolic pressure of the pathological heart sounds, have been assessed and estimated.
	(Saini et al., 2017)	N/A	The heart sound data were recorded from the healthy subjects whose age range is between 19–28 years old, whilst they were in a relaxed position, and whilst doing exercise, such as walking upstairs and downstairs through 75 steps stairs. The same recording process was then performed on the subjects whose age range is between 76–87. After the sound recording process, Hilbert-Hung Transform (HHT), which has two steps: Empirical Mode Decomposition (EMD) and Hilbert Spectral Analysis (HSA), has been used to separate heart sounds from muscle noises.
	(Nivitha Varghees & Ramachandran, 2015)	N/A	The authors have proposed a noise-robust heart sounds and murmur detection method on eGeneralMedical, Michigan heart sound and murmur library, and bioscience heart sound databases. In their proposed method, they have used two sub-bands: low frequency (LF) and high frequency (HF), to identify fundamental heart sounds (S1, S2, S3, and S4) and murmurs, respectively. Then, temporal features, including amplitude, duration, zero-crossing rate, interval, onset, and offset time-instant, have been extracted from the segmented hearts sounds and murmurs to further investigate the performance of the proposed method. The probability of correctly detecting the heart sounds is 97.33%. while the probability of correctly detecting the murmurs is 100%.
	Sharma (2015)	N/A	Hilbert transformation is applied to construct multiscale envelopes of PCG signals in order or identify the boundaries of S1 and S2 sounds. This proposed method has been assessed on heart sound datasets that are available from the Department of Medicine, Washington University. The dataset includes normal heart sounds as well as heart sounds containing various pathologies, including aortic stenosis, mitral regurgitation, pulmonic stenosis, aortic insufficiency, mitral stenosis, atrial septal defect, ventricular septal defect (YSD), and benign murmur. Besides the application of the Hilbert transformation, the authors have also performed qualitative analysis and listening tests on the segmented heart sounds to validate the obtained results from the application of Hilbert transformation.
	Sharma (2016)	N/A	Hilbert transform is used to compute a smooth envelope, and then, Teager-Kaiser Energy Operator (TKEO) is applied to identify the suitable boundaries of S1 and S2 sounds. The proposed method has been applied on a heart sound dataset that was obtained from the Department of Medicine, University of Washington. The performance of the proposed segmentation method has been evaluated by listening to the segmented heart sounds. By listening to the segmented sounds, the authors have concluded that their method can provide satisfactory results in segments S1 and S2 sounds.
	(Milani et al., 2019)	Low-pass FIR filter	The publicly available PhysioNet/CinC PCG signal dataset has been adapted in this study. First, negative frequencies have been removed from the PCG signals, whilst positive frequencies have been enveloped using an N-tap Hilbert filter to identify the prominent peaks of S1 and S2 sounds. Then, the authors have further expanded their study to extract features of segmented S1 and S2 prominent peaks, including peak to peak length, peak amplitude, area, and cycle time. These extracted features were further calibrated via Principle Component Analysis (PCA) to reduce the dimensionalities of the features before applying them to

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Table 1 (*continued*)

Segmentation method	Reference	High-frequency noise reduction	Remark
			a two-layer feedforward ANN classifier to classify normal and abnormal PCG signals. The proposed ANN classifier achieved 90% overall classification accuracy.
Shannon energy envelogram	(Saini, 2016)	N/A	The authors have recorded the heart sounds through the MP36 System of BIOPAC SYS, which has a 5000 sampling frequency and 16-bit resolution. Then, the Shannon energy envelope has been applied to obtain the positions of S1 and S2 sounds by constructing a time gate between the sound locations. This method can accurately distinguish the S1 and S2 sounds among the other present background noises and murmurs of the heart sound signals. The proposed method achieved a nearly 95.6% of success rate in detecting the locations of S1 and S2 sounds.
	Othman and Khaleel (2017)	N/A	Heart sounds have been recorded through an electronic stethoscope model 3200_3m Littman. These recordings were obtained when the participants were at different positions, including sitting, standing, etc. The recorded sounds were normalized before using the Shannon energy envelope of the input PCG signals, to identify the medium intensity of the signal and remove the noise. Then, a threshold that is based on the energy of the sound peaks has been defined to differentiate S1 and S2 sounds from murmurs. The authors have further examined the average energy envelope, an average envelope of the detected sound peaks, S1 and S2 sound durations, and present murmurs to categorize and observe the difference between the normal and abnormal sounds (abnormal sounds include the recordings of Mitral Stenosis (MS) and Mitral Regurgitation (MR) patients heart sound recordings). They have observed a significant difference between the S1, S2, and the cycle duration (S1 to next S1 or S2 to next S2), and stated that their method can be used to clearly distinguish the normal and abnormal PCG signals as well as to further examine the MS, and MR abnormal cases.
	Kang et al. (2016)	Bandpass filter with 40–500Hz frequency limits	The heart sounds used have been recorded using a 3M Littmann Model 4100 digital stethoscope. The recorded set of heart sounds have been obtained from 132 pediatric patients, including 39 clinically confirmed cases of innocent murmur, 29 cases of pathological murmurs, and another 64 normal heart sounds. However, the sound recordings were limited to a duration of 3–8 s only. The recorded sounds were then filtered before enveloping using the Shannon energy envelogram. A threshold is introduced to identify the sound lobes of S1 and S2 sounds as well as the murmurs. The proposed method has an over 99% sensitivity rate in detecting normal heart sounds and innocent murmurs. It has an 84% sensitivity rate in detecting pathological murmurs; with an overall sensitivity of 96.7%.
	Balili et al. (2016)	Sixth order Daubechies wavelet filter	The 2011 PASCAL Classifying Hearts Sounds Challenge database has been adopted in this study, and the Shannon energy envelope has been used to calculate and identify the energies of the peaks of the heart sound signal, and to separate the S1 and S2 peaks. After the S1 and S2 sound peak separation, Continuous Wavelet Transform (CWT), and Discrete Wavelet Transform (DWT) have been used to extract a total of 86 features. The extracted features were used as input to a Random Forest classifier, for the identification of murmurs, artifacts, and normal heart sounds, separately. The authors have compared the proposed method with the application of other datasets and found that their method can work well for noisy heart sound signals with high precision.
	(Salman et al., 2015)	Empirical Mode Decomposition (EMD) based de-noising method	The authors have adapted 8 heart sound signals from the University of Michigan's Heart Sound & Murmur Library. Each heart sound signal is within a duration of 70 s, and also some noises are included. They have de-noised the signals before enveloping the signal using Shannon energy envelogram, to analyse the intensities of the signal as well as to detect local maxima and minima for segmentation of S1, S2, S3, and S4 sounds.
	Hamidah et al. (2017)	Low-pass filter with 400Hz cut-off frequency	The heart sound signals were recorded through a digital stethoscope. These signals were downsampled to 2000Hz using Nyquist-Shannon's sampling theorem. The Shannon energy envelope of the downsampled signal was used to segment S1, S2, S3, and S4 sound peaks. Peak frequency, interval, duration, power,

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Table 1 (*continued*)

Segmentation method	Reference	High-frequency noise reduction	Remark
			and amplitude were calculated and used as feature inputs to detect murmurs, split and click sounds with more than 10 distinctive heart sounds.
	Banerjee et al. (2016)	Low-pass filter with 800Hz cut-off frequency	This study has used 10 PCG signals from the Peter J. Bentley Heart Sound Challenge Database, with the chosen PCG signals containing high-frequency impulsive noises. A low-pass filter was applied to remove the noises. Then the filtered signal was used to segment the S1 and S2 sound locations using a threshold that has been selected using the Shannon energy of the filtered PCG signal. This segmentation method achieves 90% accuracy.

*N/A - Not Applicable.

3. Comparison of segmentation results

The segmentation accuracies of selected research works that are based on Wavelet transform (Nivitha Varghees & Ramachandran, 2017; Golpaygani et al., 2015; Nivitha Varghees & Ramachandran, 2017; and Buza & Dizdar, 2017); Fractal decomposition (Azmy & Mohamady, 2017; E.; Koutsiana et al., 2017; Thomas et al., 2016; and; Koutsiana et al., 2017; Thomas et al., 2016); Hilbert envelope (Nivitha Varghees & Ramachandran, 2015); and Shannon energy envelogram (Kang et al., 2016; Saini, 2016; and Banerjee et al., 2016) are illustrated in Fig. 5.

As can be seen from Fig. 5, all four (4) segmentation algorithms can provide high accuracies. The highest accuracy obtained for Wavelet transform is 93.53%, Fractal decomposition is 97%, Hilbert envelope is 82.76%, and Shannon energy envelogram is 99.2%. Shannon energy envelogram gives the best overall accuracy for heart sound segmentation tasks. All four (4) segmentation methods are mathematical models, with some similarities and differences in their characteristics. Both WT and Hilbert transform split a signal into frequency bands, whilst both fractal decomposition and Shannon energy envelogram work in the time domain. WT and Hilbert transformation are the more adaptable candidates for time-frequency based signal processing applications (Banfu Yan, Ayaho Miyamoto, 2004); with Hilbert transform working well for both non-linear and non-stationary signals, whilst WT working well with non-stationary signals only (Islam Molla, Das, Hamid, & Hirose, 2013). Also, some studies have used the WT method in signal pre-processing task along with the heart sound segmentation process (Jain & Tiwari, 2018), (Patidar et al., 2015), (Nivitha Varghees & Ramachandran, 2017; Nivitha Varghees & Ramachandran, 2017). This represents the broader application of the WT method as compared to other segmentation methods described in this paper.

As compared to the other described three (3) segmentation methods, the Shannon energy envelogram is frequently used for ECG signal processing, particularly, for the evaluation of information in the QRS complex (Lee, Park, Dong, & Youn, 2018). This is because the method can diminish unwanted peaks of signals, and can be used in conjunction with other segmentation methods, such as peak intervals pattern, and threshold adaption (Lee et al., 2018). These are some of the reasons for its over 90% heart sound segmentation accuracies in different studies (Kang et al., 2016), (Saini, 2016), (Banerjee et al., 2016). Besides WT, Hilbert transforms, and Shannon energy envelogram methods, the fractal decomposition plays a major role in segmenting the low-frequency sound components, including murmurs (Hamidi et al., 2018). Murmurs may be located near high-frequency components of heart sounds, the S1 and S2 sound, and consequently, these murmurs may cause disturbance to a PCG signal, such that it could be hard to extract the real heart

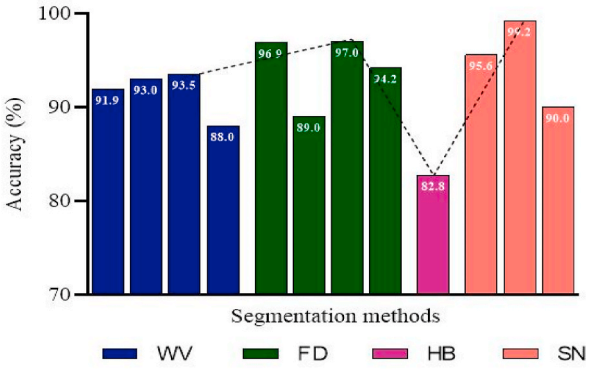


Fig. 5. Segmentation results comparison (WT-Wavelet transform (Nivitha Varghees & Ramachandran, 2017), (Golpaygani et al., 2015), (Nivitha Varghees & Ramachandran, 2017), (Buza & Dizdar, 2017), FD-Fractal decomposition (Azmy & Mohamady, 2017), (E. Koutsiana et al., 2017), (Koutsiana et al., 2017), (Thomas et al., 2016), HB-Hilbert envelope (Nivitha Varghees & Ramachandran, 2015), SN- Shannon energy envelogram (Kang et al., 2016), (Saini, 2016), Banerjee et al., 2016)).

sounds. However, murmurs act very similar to fractals, such that fractal decomposition may be able to separate them effectively (Hamidi et al., 2018). Consequently, fractal decomposition-based heart sound segmentation methods have been used in several studies (Azmy & Mohamady, 2017), (Koutsiana et al., 2017), (Koutsiana et al., 2017), and (Thomas et al., 2016), and have been demonstrated to give high accuracies. Overall, these four (4) general segmentation methods are priceless tools that can help unravel the complexities of a heart sound signal.

The WT and fractal decomposition are known to be spectro-temporal, looking into the frequency properties of a signal. Hilbert transform has been applied into multiple applications, including amplitude, phase or frequency modulation analysis, while Shannon energy estimates the intensities of signal components. Among these methods, WT can naturally conduct signal analysis and signal splitting into different scale components with the adoption of Fourier basis (Cattani, 2008), whilst giving a good overview of the spectral changes in power over time. Thus, it can balance both temporal and frequency precisions. Mainly, WT is best used to characterize temporal changes of spectral subtleties of a signal. Hilbert transform may also be used to identify prominent frequencies based on their high power, however, as compared to WT, Hilbert transform can estimate the understated changes in the frequencies (Cohen & Cohen, 2014). Comparison between Hilbert transform and envelop analysis, it is very difficult to comprehend a wide band signal using Hilbert transform, with the amplitude decomposition not providing sufficient performance to allow analysis. On the other hand, envelop analysis can provide a good amplitude analysis of a signal (Flandrin, Rilling, & Gonçalvés, 2004). Nevertheless, in the analysis of ECG signal, both Hilbert transform and Shannon energy envelope can detect the R-peaks, but both contains many delays and use a high memory, and hence may not be well suited for a real time signal detection (Beyramienanlou & Lotfivand, 2017). Shannon energy envelope is a simple way to convert the signal energy into an energy package, which can be used to thoroughly investigate the spikes and the peaks of a signal component.

The adaption of two (2) segmentation methods also can provide good results in heart signal analysis, where (Zidelmal, Amirou, Ould-Abdeslam, Moukadem, & Dieterlen, 2014) created a frequency dependent regulation using the Shannon energy and S-transform; which include both short time Fourier transform (STFT) and WT. Similar to the WT method, fractal decomposition method can also be applied into non-stationary signals to obtain the time-frequency information. Fractal decomposition can represent a signal complexity in the time domain through a sliding window; thus; it gives a point to point information of the fractal signal (E. Koutsiana et al., 2017). Also, fractal decomposition can provide better segmentation results irrespective of the complexity of a signal, hence it is capable of extracting the features to measure fractal dimensions (OK & Rajesh, 2020).

Apart from these four (4) popular methods, the study of (Papadaniil & Hadjileontiadis, 2014) has implemented Ensemble Empirical Mode Decomposition (EEMD) to detect the locations of S1 and S2 sounds and achieved 94.56% segmentation accuracy. On the other hand, Nieblas, Alonso, Conte, and Villarreal (2013) have applied the Matching pursuit algorithm to segment S1 and S2 sounds, to identify the cardiac cycle events onset and offset from PCG signals. Segmentation accuracies of 97.5% and 96% for the onset and offset, respectively, have been shown using the method. Likewise, Springer, Tarassenko, and Clifford (2016) have segmented S1 and S2 sounds using the Hidden semi-Markov model and achieved an average F1 score of $95.63 \pm 0.85\%$%, whilst Sedighian, Subudhi, Scalzo, and Asgari (2014) have also used the same segmentation model, to achieve $92.4\% \pm 1.1\%$ and $93.5\% \pm 1.1\%$ accuracy in segmenting S1 and S2 sounds, respectively. (Santos's', 2001) has introduced a heart sound segmentation method, which is based on time-frequency analysis with a combination of threshold values. The threshold values are defined using Rényi entropy to identify the amplitudes of main heart sounds and murmurs. This method has been shown to achieve a 0.8% of error ratio. These heart sound segmentation results show that unpopular methods also can provide good accuracies similar to the popular methods.

Deep learning-based heart sound segmentation methods are also another trending technology that has been used in many research studies (T. E. Chen et al., 2017), (Messner, Zöhrer, & Pernkopf, 2018), (Fernando et al., 2020), (Renna, Oliveira, & Coimbra, 2019), and (Y. Chen, Sun, Lv, Jia, & Huang, 2021). Chen et al. (2017) have converted heart sound signals into a sequence of Mel-frequency cepstral coefficients (MFCCs) and adapted a K-mean algorithm to refine and cluster the MFCCs into two groups by considering their discriminative capabilities and depiction. Deep Neural Network (DNN) is then used on these refined MFCCs to identify S1 and S2 sounds. The method has been shown to achieve a 91.12% accuracy rate, with the method providing better performance than KNN, GMM, Linear Regression (LR), and SVM methods. Messner et al. (2018) have divided the PCG signals into frames to facilTitate labelling of S1, S2, systole, and diastole intervals through hand-corrected annotation by observing the 20ms shift time of the frames. The authors have also proposed a general solution to detect events that occurs in systole and diastole intervals, including extra heart sounds (S3 and S4), murmurs, and other acoustic events, using multi-label Deep Recurrent Neural Network (DRNN) with a softmax output layer. The proposed method has successfully achieved an event-based F-score of 95.6% on an independent heart sound test set. However, the authors later compared the achieved results with the logistic regression hidden semi-Markov model based heart sound segmentation method and have found that the F-score is almost the same as the DRNN method. Similarly, Fernando et al. (2020) have proposed a novel method to segment S1 and S2 heart sounds and systole/diastole intervals of a PCG signal. In their study, a combination of DNN and attention-based learning framework have been adopted to achieve over 90% accuracy. Deep Convolutional Neural Networks (DCNN) is another deep learning-based approach that can be used in the heart sound signal segmentation process. Renna et al. (2019) have used DCNN to segment the main heart sounds including S1 and S2 sounds of a PCG signal to give an average sensitivity of 94%. Likewise, Chen, Sun, Lv, et al. (2021) have also utilised Deep Convolutional Recurrent Network (DCRN) to detect S1 and S2 sounds as well as the systole and diastole intervals of a PCG signal. The authors compared their proposed method with some other competitive methods, including Linear Regression based Hidden semi-Markov model (LR-HSMM), and CNN, and have concluded that the proposed method is capable of achieving similar segmentation results to its competitors.

A few studies have also reviewed some of the present heart sound segmentation approaches. Among those studies (Li et al., 2020), and (Nabih-Ali, El-Dahshan, & Yahia, 2017) have provided an in-depth discussion on the work that has focused on Hidden Markov Models (HMM), WT, correlation coefficient matrices, and deep learning-based approaches. It is noted that some of the deep learning

approaches, such as DNN, CNN, and RNN that are popular in the present have shown relatively lower heart sound segmentation performance as compared to the older methods (Li et al., 2020). However, deep learning approach works particularly well in heart sound classification task (W. Chen, Sun, Lv, et al., 2021), specifically in S1, S2, S3, and S4 heart sound recognition (T. E. Chen et al., 2017). Deep learning based algorithms require a proper training process and good set of training data; which can be used to define the characteristics of a segmented heart sound (i.e., S1, S2, S3, and S4). Then (C. Liu et al., 2016) have discussed different heart sound databases, including The Massachusetts Institute of Technology heart sounds database, The Aristotle University of Thessaloniki heart sounds database, and The K N Toosi University of Technology heart sounds database, as well as providing a very brief review of the different heart sound segmentation methods, including envelope-based methods, feature-based methods, machine learning methods, and HMM models. They have further described the applications of these methods with reference to many other past research studies, such as (Jiang & Choi, 2006), (Castro, Vinhoza, Mattos, & Coimbra, 2013), and (Sun, Jiang, et al., 2014). However, most of the publicly available datasets relevant to heart sound segmentation methods consist of small datasets only (Koutsiana et al., 2017), (Thomas et al., 2016). Generally, deep learning methods can only provide a good solution for a large database (Li et al., 2020), for proper training of the model.

The older methods such as; WT, Fractal decomposition, Hilbert envelope, and Shannon energy envelogram are uncommonly used in the heart sound classification process, while deep learning methods are used in both segmentation and classification tasks. This may be the most crucial disadvantage of using the older heart sound segmentation methods over the deep learning based methods. Nevertheless, the older methods can breakdown the heart sounds into segments to extract their information for further processing; including the heart sound classification and disease recognition, that provides higher accuracies in the heart sound classification (Milani et al., 2021). The older methods particularly work well in locating the heart sounds and their boundaries or providing a smooth enveloped signal (Sharma, 2015). In their study, (Milani et al., 2021) have applied Hilbert envelope to segment S1 and S2 sounds, then the segmented sounds are used to classify the normal and abnormal heart sounds using a supervised machine learning approach to give over 90% classification accuracy. This demonstrates that a combined segmentation method with machine learning approach can provide even better performance in heart sound classification.

The enveloped based heart sound extraction is another form of application that has been recently used in (Kamson, Sharma, & Dandapat, 2020). The sigmoid characteristics of the logistic function is adopted by (Kamson et al., 2020), to separate the S1 and S2 heart sounds along with some silent or noise interfered systolic and diastolic intervals. This method is named as logistic function based amplitude moderation (LFAM) envelogram method, which considers the amplitudes of the enveloped signals to identify and extract the information of fundamental heart sounds and systolic/diastolic intervals. Then the extracted information is further evaluated by HSMM as a classifier and achieved an average F1 score of 97.73%. It is evident that the combination of separate segmentation methods, especially the older segmentation methods with machine learning or deep learning algorithms are common in many applications (Balili et al., 2016), (Milani et al., 2019), and they have provided better results for their purpose of study; thus, it can be noted that, even the older heart sound segmentation algorithms can still be used in the present for any heart sound related applications. Many researchers have demonstrated superior results on the four (4) segmentation methods as compared to deep learning approaches in segmentation work. Indeed (Li et al., 2020), have found that fractal decomposition, wavelet transform, and envelope-based techniques can provide over 90% accuracy in segmenting different heart sounds and murmurs.

4. Conclusion

This paper reviews the existing approaches used in heart sound segmentation methods that can be used for segmenting PCG and ECG signals. Specifically, four (4) heart sound segmentation methods have been discussed: wavelet transforms, fractal decomposition, Hilbert envelope, and Shannon energy envelogram. These selected time-frequency domain heart sound segmentation methods have been used to develop many heart sound signal processing applications. For instance, wavelet transformation and Hilbert enveloping methods have been implemented in some research studies to differentiate fundamental heart sounds and murmurs.

As is evident from the literature, more research works have used wavelet transform and Hilbert enveloping methods, than other methods for segmentation tasks. These two methods can act as a bandpass filter to de-noise the high-frequency noises and can be used to analyse the signals using different resolutions. Using low-pass or bandpass filters to de-noise heart sound signals is very popular in recent research studies, with these types of filters capable of elaborating the most critical information of the heart sound signals whilst ignoring the rest. Thus, the combination of these filters with fractal decomposition and Shannon energy envelogram have been shown to provide the highest segmentation accuracies as compared to purely using the wavelet transform and Hilbert envelope methods only.

All four (4) segmentation methods have been shown to perform well for any non-linear and non-stationary PCG and ECG signals in identifying the fundamental heart sound and extracting their instantaneous frequencies. These definitely assist in the development of an effective heart sound detection and classification algorithm.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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